Layer 2 v0.2.1 — `team_width` Warning Review Plan

Purpose

The goal is to decide whether `team_width` is safe enough for **Layer 3 descriptive tactical profiling**, whether it should be used only with flags/cautions, or whether it needs to be excluded or patched before downstream use.

This is a review plan only. No Layer 3 profiling starts yet.

The Layer 2 handoff says the football sanity workflow completed successfully, but the final QA status was **warning** because `team_width_pitch_bound` exceeded the current pitch-width reasonableness bound at least once. It also says this should be reviewed before using `team_width` for stronger tactical claims.

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1. What exactly should be inspected?

Inspect five things:

A. The warning itself

Identify the exact reason `team_width_pitch_bound` triggered:

- How many rows exceeded the pitch-width bound?
- How far above the bound were they?
- Were the exceedances tiny tolerance issues or major impossible values?
- Were they concentrated in one match, one team, one phase, or one frame range?

B. The outlier rows

For every row where `team_width` exceeds the expected pitch-width reasonableness bound, inspect:

- `match_id`
- `frame`
- `period`

- `team_id`
- `phase_index`
- possession status
- in-possession phase label
- out-of-possession phase label
- `team_width`
- `team_depth`
- `convex_hull_area`
- `avg_distance_to_centroid`
- `player_count_used`
- `detected_player_count`
- `extrapolated_player_count`
- `metric_warning_flag`

The goal is to determine whether the warning reflects real extreme team spread, coordinate overflow, extrapolated tracking noise, a frame-level artifact, or a metric definition issue.

C. The raw player coordinates behind the outliers

For each flagged frame-team row, inspect the player-level coordinates that produced the width:

- min `y_metric`
- max `y_metric`
- players causing the min/max values
- whether those players were detected or extrapolated
- whether coordinates fall outside expected pitch bounds
- whether the team has the expected number of players
- whether one player is causing the whole issue

This matters because `team_width` is defined as max `y_metric` minus min `y_metric`, so one bad/extreme player coordinate can trigger the warning.

D. Visual plausibility

Review the generated pitch plots and, if necessary, request new plots for the exact flagged frames.

Check whether:

- players appear inside or near pitch bounds
- team separation looks plausible
- the wide team shape looks like a real football shape
- one player is wrongly projected far outside the pitch

- convex hull shape is sensible
- the ball/team/phase context looks plausible

The handoff says pitch-shape validation plots should show pitch outline, player locations, centroids, hulls when valid, ball location, match/frame/period/phase label, and team-in-possession context.

E. Time continuity

Inspect phase-segment or frame-neighborhood time series around the outlier:

- Does `team_width` spike for one frame and immediately return to normal?
- Does it stay high over a plausible stretch of play?
- Does the spike coincide with many extrapolated players?
- Do related metrics also spike, especially `convex_hull_area` or `avg_distance_to_centroid`?

A one-frame spike is more likely a tracking artifact. A sustained high value may be valid, especially if visually plausible.

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2. Which files should be used?

Primary files:

1. `data/processed/features/layer2_v02_football_sanity_summary.json`
   Use this to confirm the warning status, thresholds if recorded, and summary-level reasonableness results.
2. `data/processed/features/layer2_v02_football_sanity_summary.csv`
   Use this to inspect metric summaries by phase and identify where the warning appears.
3. `data/processed/features/frame_team_shape_metrics.csv`
   This is the main table for row-level inspection because it contains frame-level `team_width`, phase context, and quality-control fields.
4. `data/processed/features/phase_segment_summaries.csv`
   Use this to check whether problematic `team_width` behavior persists at the phase-segment level or is only a frame-level issue.
5. `outputs/figures/layer2_sanity/`
   Use existing pitch plots and phase-segment time-series plots. The handoff says 11 validation plots were created there.

Secondary files if needed:

6. `data/processed/skillcorner/{match_id}/canonical_tracking_players.csv`
   Use this to inspect the player coordinates and detection flags behind flagged frame-team rows.
7. `data/processed/skillcorner/{match_id}/match_metadata.json`
   Use this to confirm pitch width/length for the specific match.
8. `src/soccer_tips/features/team_shape.py`
   Only inspect the definition if the row/plot review suggests a possible width/depth semantic mismatch or implementation bug. Any code patch should go to the Lead Developer.

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3. Which output rows or plots should be checked?

Check these row groups:

Row group 1: All pitch-bound violations

Every row where:

- `team_width` exceeds the accepted pitch-width bound used in football sanity QA.

These are the decisive rows.

Row group 2: Top largest `team_width` values

Even if only some triggered the warning, inspect the top values:

- top 20 largest `team_width` rows overall
- top 10 by match
- top 10 by phase label, if phase-linked

This checks whether the warning is isolated or part of a wider distribution problem.

Row group 3: Neighboring frames around flagged rows

For each serious outlier, inspect a frame window around it:

- same match
- same team
- same period
- frames around the outlier, for example before/after the flagged frame

This determines whether the value is a one-frame artifact or sustained behavior.

Row group 4: Phase-segment summaries containing flagged frames

Check whether flagged frame-level values meaningfully affect:

- phase-segment mean `team_width`
- phase-segment max `team_width`
- phase-segment standard deviation
- phase-segment sample count

If a few outlier frames distort segment-level summaries, Layer 3 may need filtering or robust aggregation.

Row group 5: Existing validation plots

Check:

- pitch-shape sample plots
- phase-segment time-series plots

If the existing plots do not include the flagged frames, ask the Lead Developer or developer helper to generate exact outlier-frame plots.

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4. Decision rules for `team_width`

A. Classify as valid

Use this only if:

- outlier values are not actually impossible after considering pitch tolerance
- player coordinates look plausible on pitch plots
- no obvious player is wrongly projected far outside the pitch
- detection/extrapolation counts are acceptable
- width/depth convention is confirmed: width = lateral `y_metric` range, depth = longitudinal `x_metric` range
- values are stable over nearby frames when large
- phase-segment summaries are not distorted by isolated spikes

Result: `team_width` can enter Layer 3 descriptive profiling.

Still use careful language. It should support descriptive structural width claims, not final tactical identity claims.

B. Classify as valid with outlier flags

Use this if:

- most `team_width` values are plausible
- violations are rare
- violations are explainable
- plots mostly look football-realistic
- outliers are small or concentrated in known noisy situations
- keeping the metric is useful, but extreme rows need traceability

Result: `team_width` can enter Layer 3, but Layer 3 outputs should carry a `team_width_warning_flag` or similar quality flag.

C. Classify as valid with filtering/tolerance rules

Use this if:

- the metric definition is correct
- most values are useful
- a small number of rows are clearly artifacts
- artifacts are identifiable by rules such as coordinate overflow, extreme extrapolation count, or one-frame spikes
- filtering does not remove meaningful football behavior

Result: `team_width` can enter Layer 3 only after the Lead Developer approves the filtering/tolerance rule.

Important: filtering rules are implementation logic, so the Lead Developer owns the final code-level decision.

D. Classify as unsafe for Layer 3

Use this if:

- outliers are frequent
- outliers are large and not visually plausible
- values are unstable across adjacent frames
- extreme values materially distort phase-segment or match-level summaries
- the problem appears across multiple matches/teams/phases

- detection/extrapolation issues make the metric unreliable

Result: exclude `team_width` from Layer 3 v0.1 tactical claims. It may remain in outputs as a raw/flagged diagnostic field, but not as evidence for width, compactness, or tactical identity.

E. Classify as requiring code/definition patch

Use this if:

- `team_width` appears to be using the wrong axis
- width/depth convention is reversed
- pitch dimensions are being applied incorrectly
- metric coordinates are mismatched with raw coordinates
- pitch-width bounds are hardcoded incorrectly
- rows show impossible values that cannot be explained by tracking noise
- the code definition conflicts with the project's football convention

Result: stop downstream use of `team_width` and send the issue to the Lead Developer for patch/review before Layer 3 uses it.

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5. Evidence enough to allow `team_width` into Layer 3 descriptive profiling

Enough evidence would be:

- A count and percentage of violating rows showing the issue is rare or explainable.
- A table of the worst outlier rows with match/team/frame/phase/quality fields.
- Visual review of flagged examples showing plausible or explainable player positioning.
- Confirmation that `team_width = max(y_metric) - min(y_metric)` matches the project convention.
- Confirmation that pitch dimensions are being interpreted correctly for SkillCorner.
- Confirmation that outliers do not materially distort phase-segment or match-level summaries.
- A documented decision: use normally, use with flags, or use with filtering/tolerance.

If all of that is satisfied, `team_width` can be used in Layer 3 v0.1 as a descriptive metric, with documented caution if needed.

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6. Evidence that means `team_width` should be excluded or flagged

Exclude or flag `team_width` if any of these are true:

- violations are not rare
- violations are much larger than reasonable pitch-width tolerance
- flagged rows are driven by players far outside the pitch
- flagged rows are driven by extrapolated players
- values spike for isolated frames without football explanation
- phase-segment summaries are distorted by frame-level artifacts
- pitch plots show impossible shapes
- width/depth axis convention is uncertain
- the metric definition or pitch-bound logic needs a patch
- the review cannot explain the warning clearly

Default analytical stance: if unresolved, `team_width` should not support Layer 3 tactical claims. It can be carried as a flagged metric if useful.

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7. What should be sent to the Lead Developer after the review?

Send a short review package with:

1. **Status decision**
    - valid
    - valid with outlier flags
    - valid with filtering/tolerance rules
    - unsafe for Layer 3
    - requires code/definition patch
2. **Summary of the warning**
    - number of violating rows
    - percentage of total frame-team rows
    - max `team_width`
    - typical/median `team_width`
    - affected matches, teams, periods, and phases
3. **Worst examples table**
    - match_id
    - frame
    - period

- team_id
- phase labels
- possession status
- team_width
- player_count_used
- detected_player_count
- extrapolated_player_count
- notes

4. **Visual evidence**
   - list of reviewed plots
   - whether flagged frames looked plausible or corrupted
   - whether new exact outlier plots are needed
5. **Cause assessment**
   - true edge/outlier behavior
   - tolerance issue
   - broadcast/extrapolated tracking artifact
   - aggregation issue
   - width/depth semantic mismatch
   - code/definition issue
   - unresolved
6. **Layer 3 recommendation**
   - allow `team_width`
   - allow with flags
   - allow only after filtering
   - exclude from Layer 3 v0.1
   - patch before use
7. **Implementation request, if any**
   - no code change needed
   - add warning flag
   - add filtering/tolerance rule
   - adjust sanity threshold
   - patch metric definition
   - generate additional validation plots

Recommended controlled review sequence

1. Read `layer2_v02_football_sanity_summary.json`.
2. Identify the exact `team_width_pitch_bound` warning details.
3. Pull all violating `frame_team_shape_metrics.csv` rows.
4. Rank the largest `team_width` rows.
5. Inspect raw player coordinates for the worst examples.

6. Review existing pitch and time-series plots.
7. If existing plots do not show the flagged frames, request exact outlier-frame plots.
8. Check whether outliers distort `phase_segment_summaries.csv`.
9. Classify the metric using the decision rules above.
10. Send the Lead Developer the review package before Layer 3 begins.

My recommendation: do this review before defining the Layer 3 tactical fingerprint schema, because the fingerprint schema depends on whether `team_width` is trusted, flagged, filtered, or excluded.